A Constructive Introduction to Quantum Field Theory

Φ^4_3 on the torus I: the FBSDE in force coordinates and renormalisation

Massimiliano Gubinelli | University of Oxford

Joint work with Sarah-Jean Meyer, with the use of AI

SISSA lectures, Lecture 5 (provisional) — notes: Lecture 5, §§10–12

Where we are

Lecture 4. Interacting field at resolution t = solution of the FBSDE

$$\Phi_t = \int_0^t \dot{Q}_s F_s \, \mathrm{d}s + \int_0^t \dot{Q}_s^{1/2} \mathrm{d}W_s, \qquad F_t = \mathbb{E}_t\left[-D\,V_T(\Phi_T)\right].$$

In the **forecast coordinate** $\psi_t = \mathbb{E}_t \Phi_T$ the forward equation has no drift and the trees are integrated exactly; the backward half is the **force equation**.

Lectures 5–6. Construct Φ^4_3 on $\mathbb{T}^3$ with this FBSDE:

1. **deterministic renormalisation**: an explicit approximate force $\bar{\mathscr{F}}_q$ and its defect (today);
2. **a priori estimates** for the exact FBSDE in the coordinates of $\bar{\mathscr{F}}_q$, uniform in the lattice (Lecture 6);
3. **removing the cutoffs** and identifying the limit (Lecture 6).

G., course notes on Φ^4_3 via FBSDEs (2026); Bailleul–Chevyrev–G. (in prep.)

What is proved

For every coupling $\lambda > 0$ and every mass $m^2 = q_0 > 0$, on a fixed torus, with no smallness, and an explicit counterterm with arbitrary finite part:

- the whole lattice sequence converges to a measure ν, an explicit density over an explicit SDE;
- the forecasts converge to the solution of an FBSDE from $\Upsilon_{q_0} = 0$, driven by the limit of the lattice innovations, and $\psi_q = m_q(\Phi_q)$;
- uniqueness in law for its scalar form, a variational formula with no infinite constant, quartic tails, integration by parts.

Not proved: the vector FBSDE alone, strong solutions, OS axioms in full, infinite volume.

Guiding principle. Do not estimate Φ and F. Estimate the error of an explicit approximate forecast.

The results of Lectures 5–6 are new: joint work with S.-J. Meyer + AI.

The problem and the exact FBSDE

Lecture 5

Lattice and the mass-regulator profile

$\Lambda_N = (N^{-1}\mathbb{Z}/\mathbb{Z})^3$, $\int f = N^{-3}\sum_x f(x)$, $\Delta = \Delta_N$ nearest neighbour. Resolution parameter $q = q_0 + t^2 \in [q_0, \infty)$:

$$K_q = q - \Delta, \qquad C_q = K_q^{-1}, \qquad C = C_{q_0} = (m^2 - \Delta)^{-1}.$$

In Lecture 3 notation ($T = \infty$, and $\dot{Q} = \partial_q Q_q$):

$$Q_q = C - C_q, \quad \dot{Q} = C_q^2, \quad \dot{Q}^{1/2} = C_q, \quad S_q = C_q, \quad \mathscr{R}_q = q - q_0.$$

Free resolvents (uniform in N)

$C_q \geqslant 0$ as a kernel, $C_q 1 = q^{-1} 1$, $\|C_q\|_{p\to p} = q^{-1}$,

$$a_2(q) = (C_q^2)(x,x) \lesssim q^{-1/2} \wedge N^3 q^{-2}, \qquad \mathrm{Tr} C \asymp N.$$

The **bubble** $a_2(q)$ is the coincident-point variance of the shell at resolution q.

The cutoff measure

$U(\varphi) = \frac{\lambda}{4}\int \varphi^4$, $V_N = U + \frac{\mu_N}{2}\int \varphi^2$, $\nu_N(\mathrm{d}\varphi) = \mathcal{Z}_N^{-1} \mathrm{e}^{-V_N(\varphi)} \mu^C(\mathrm{d}\varphi)$.

$\gamma = 3\lambda$, so $D^2 U(y) = \gamma y^2$ (not the counterterm γ_T of Lecture 4). The Wick ordering is now part of μ_N.

Main result, informal. Let $\mu_N = r - 3\lambda \operatorname{Tr} C + 6\lambda^2 \int C^3$. Then (a) $\nu_N \to \nu$ in H^{-s}, $s > \frac{1}{2}$, along the whole sequence; (b) the forecasts converge in law to a solution of a limiting FBSDE; (c) ν has a variational formula, an integration by parts formula and quartic tails.

These come from bounds on the error Z of an approximate forecast, uniform in N: **UV stability**.

The Polchinski process as a filter

By Lecture 4, in the profile $S_q = C_q$, $\mathscr{R}_q = q - q_0$; at fixed N everything is a theorem. The Polchinski process solves

$$\mathrm{d}\Phi_q = C_q^2 F_q \,\mathrm{d}q + C_q \,\mathrm{d}W_q, \quad F_q = \mathbb{E}_q[-D\,V_N(\Phi_\infty)], \quad \Phi_{q_0} = 0,$$

and $\Phi_q \to \Phi_\infty \sim \nu_N$.

- a noisy observation of its endpoint: $\Phi_q = (q - q_0) C_q \Phi_\infty + C_q B_{q-q_0}$, B independent of Φ_∞;
- forecast $\psi_q = \mathbb{E}_q \Phi_\infty = \Phi_q + C_q F_q = m_q(\Phi_q)$, covariance $\mathsf{Cov}_q > 0$;
- W is the innovation: $\mathrm{d}\psi_q = \mathsf{Cov}_q \,\mathrm{d}W_q$.

One site

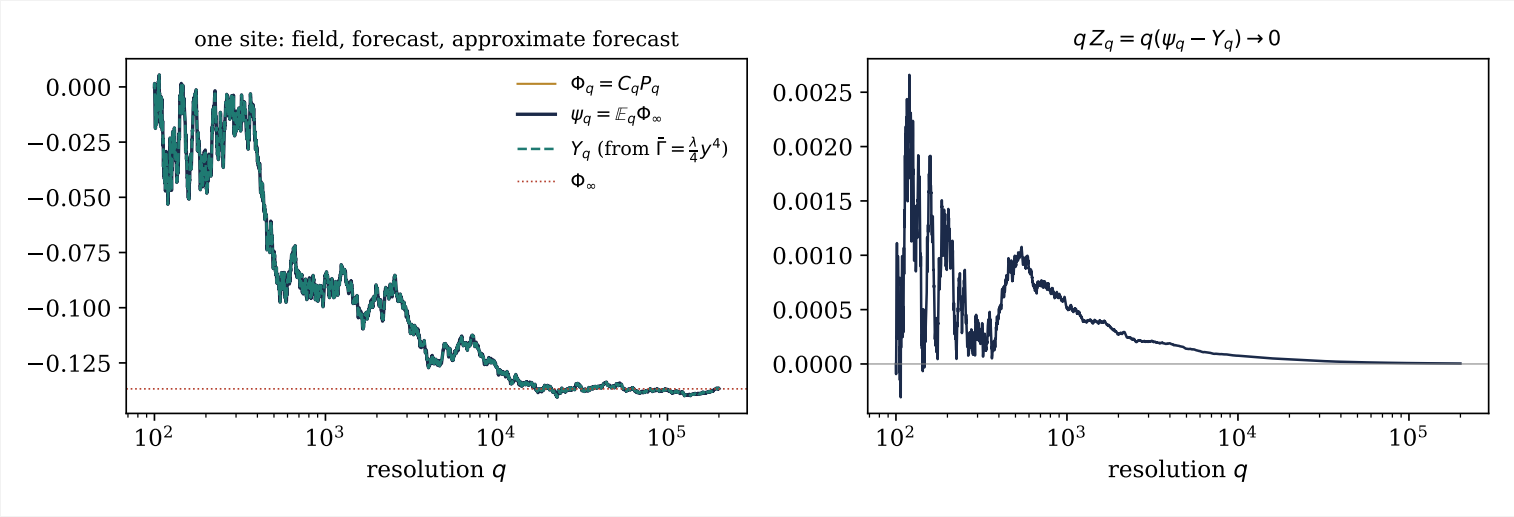


Illustration: $\nu \propto e^{-\lambda\varphi^4/4}\mu^{1/q_0}$ on one site, $\lambda = 2$, $q_0 = 100$ ($g_{q_0} = 0.6$). Φ_q, the posterior mean ψ_q, and the approximate forecast Y_q from $\bar{\mathscr{F}} = \lambda y^3$. All converge to the hidden Φ_∞, and $qZ_q \to 0$.

The exact force in the forecast coordinate

Since $\Phi_q \mapsto \psi_q$ is invertible, define $\mathscr{F}_q$ by $\Phi_q = \psi_q + C_q \mathscr{F}_q(\psi_q)$ (the field equation $K_q \Phi_q = K_q \psi_q + \mathscr{F}_q(\psi_q)$), and $H_q^{\mathrm{ex}}(y) = (K_q + D\mathscr{F}_q(y))^{-1}$.

It is the $\mathscr{F}$ of Lecture 4 ($x = \psi + S\mathscr{F}(\psi)$), and $F_q = -\mathscr{F}_q(\psi_q)$.

The force equation with a mass regulator

$$\partial_q \mathscr{F}_q + \frac{1}{2} \mathrm{Ctr}_{(H_q^{\mathrm{ex}})^2} D^2 \mathscr{F}_q = 0, \quad \mathrm{Cov}_q = H_q^{\mathrm{ex}}(\psi_q).$$

Itô on $\Phi = \psi + C\mathscr{F}(\psi)$. Its integrated form is Wetterich's equation, which we do not need. An identity about the unknown ν_N, not a construction. Strategy: an approximate force $\bar{\mathscr{F}}$ gives approximate forecast coordinates; the error should be driven by the failure of $\bar{\mathscr{F}}$ to solve this equation.

Approximate forecast and error

$\bar{\mathscr{F}}_q$ odd, $D\bar{\mathscr{F}}_q$ symmetric, $K_q + D\bar{\mathscr{F}}_q \geqslant \bar{c}_q$.

$$H_q = (K_q + D\bar{\mathscr{F}}_q)^{-1}, \quad \mathscr{D}_q = \partial_q \bar{\mathscr{F}}_q + \frac{1}{2}\mathrm{C}\mathrm{tr}_{H_q^2} D^2 \bar{\mathscr{F}}_q, \quad b_q = -H_q \mathscr{D}_q.$$

$$\Upsilon_q + C_q \bar{\mathscr{F}}_q(\Upsilon_q) = \Phi_q, \quad Z_q = \psi_q - \Upsilon_q, \quad \Xi_q = \mathrm{Cov}_q - H_q(\Upsilon_q).$$

Exact, at fixed N

$$\mathrm{d}\Upsilon_q = [b_q(\Upsilon_q) + H_q(\Upsilon_q) Z_q]\,\mathrm{d}q + H_q(\Upsilon_q)\,\mathrm{d}W_q,$$

$$\mathrm{d}Z_q = -[b_q(\Upsilon_q) + H_q(\Upsilon_q) Z_q]\,\mathrm{d}q + \Xi_q\,\mathrm{d}W_q, \quad \Upsilon_{q_0} = Z_{q_0} = 0.$$

Proof: Itô on the coordinate equation, with $\mathrm{d}P = \psi\,\mathrm{d}q + \mathrm{d}W$: the martingale part gives H, the drift gives $HZ + b$.

Reading the system

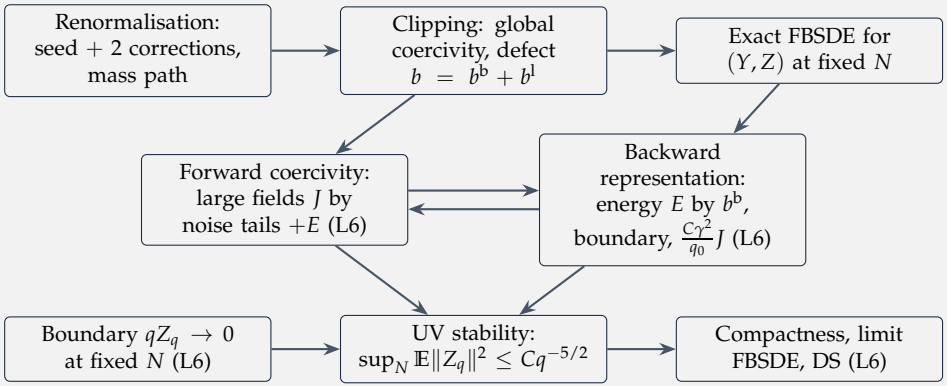
- Y: forward and explicit once Z is known; coefficients b, H built from $\bar{\mathscr{F}}$.
- Z: backward. The terminal condition $q\,Z_q \to 0$ is **proved** at fixed N (Lecture 6), because $\bar{\mathscr{F}}_q \to$ bare force.
- The only source of Z is the drift b, the force defect, transported by $\exp \int H$.

UV stability

The explicit force $\bar{\mathscr{F}}$ computes the forecast up to an error Z which is small uniformly in N.

Needed: (i) a defect integrable against the backward propagator (**renormalisation**); (ii) global coercivity of $\bar{\mathscr{F}}$ (**large fields**); (iii) a priori estimates (Lecture 6).

The structure of the construction



Renormalisation: why two of them

Lecture 5

Scales and power counting

$$g_q = \gamma\, q^{-1/2} (\text{dimensionless coupling}),$$

$$r_q = (q/\gamma)^{1/2} (\gamma\, r_q^2 = q : \text{nonlinearity} \approx \text{regulator}).$$

Free field at resolution q: $\mathrm{Var} X_q(x) \lesssim q^{1/2}$, typical size $q^{1/4} = g_q^{1/2} r_q \ll r_q$.

Rooted norm of a vertex $T(x_1, \dots, x_k)$ and field ball:

$$s(T) = \max_i \sup_{x_i} \int |T| \prod_{j \neq i} \mathrm{d}x_j, \qquad \mathsf{S}_k(f) = \sup_{\mathscr{B}_q} s\,(D^k f), \qquad \mathscr{B}_q = \{\|y\|_\infty < 2\, r_q\}.$$

Natural size of the kth vertex of the **loop corrections**:

$$\mathsf{S}_k \lesssim g_q\, q\, r_q^{2-k} \qquad (k = 4 : \gamma\, g_q).$$

The loop estimate

Loop estimate

A closed loop with p vertices: $s(\text{loop}) \leqslant C a_2(q)\, q^{-(p-1)} \prod_i s(T_i)$ (two propagators in L^2, the others in operator norm). After the q integration a one-loop term is of size $g_q q r_q^{2-k}$; every further loop gains a factor g_q.

The seed and the corrections

Seed (one loop): replace $D\mathscr{F}$ by γy^2 and integrate from $q=\infty$, with the linear part subtracted,

$$\ell_q(y)(x)=\gamma y(x)\,[G_q(y)(x,x)-C_q(x,x)\,],\quad G_q(y)=(K_q+\gamma y^2)^{-1}.$$

Leading term = bubble, $s(D^3\ell_q(0))=6\gamma^2 a_2(q)$.

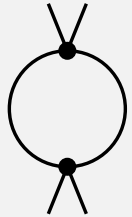
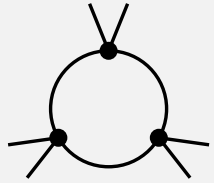
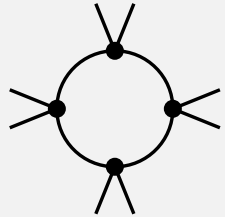
Corrections. Π removes the linear jet, $M_q=\mu_q+\Theta_q$ a linear path, $U'=\lambda y^3$,

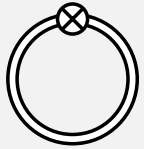
$$\mathscr{T}_q[M,\ell\,]=-\frac{1}{2}\mathrm{C}\mathrm{tr}_{R^2}D^2(U'+\ell\,),\quad R=(K_q+M_q+D(U'+\ell\,))^{-1},$$

$$\ell^{[0]}=\ell\,,\quad \ell_q^{[j+1]}=\ell_q-\int_q^\infty \Pi(\mathscr{T}_s[M,\ell^{[j]}]-\partial_s\ell_s)\,\mathrm{d}s,\quad j=0,1,$$

$$\mathscr{F}_q^{\,o}=U'+M_q y+\ell_q^{[2]}\quad \text{on the ball.}$$

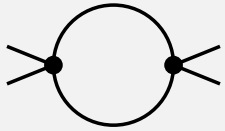
Seed and corrections in graphs

$$\ell_q(y) \;=\; D\,\Big[-\tfrac{1}{4}$$  $$+\tfrac{1}{6}$$  $$-\tfrac{1}{8}$$  $$+\;\cdots\;\Big]$$

$$\partial_q \ell_q^{[j+1]} \;=\; -\Pi\;\tfrac{1}{2}$$  $$\ell[j]$$

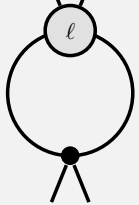
double line: $(K_q + M_q + D(U' + \ell^{[j]}))^{-1}$
cross: $\partial_q \mathcal{R} = 1$

seed $\ell^{[0]} = \ell$
one loop, g_q



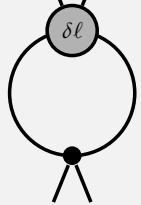
$\longrightarrow$

$\ell^{[1]} - \ell^{[0]}$
two loops, g_q^2



$\longrightarrow$

$\ell^{[2]} - \ell^{[1]}$
three loops, g_q^3



$\Longrightarrow$

stop: defect
$e^{\circ} = \mathcal{O}(g_q^3)$
integrable source

Each step inserts the previous correction into the loop: one more loop, one more g_q. Two steps give a defect of order g_q^3.

The mass

The linear jet is not iterated: it is solved exactly by M,

$$\partial_q M_q = -\gamma \mathrm{Tr}(R_{0,q}^2) I - \frac{1}{2} C \mathrm{tr}_{R_{0,q}^2} D^3 \ell_q^{[2]}(0), \quad R_{0,q} = (K_q + M_q)^{-1},$$

with $\mu_{q_*} = b$, $|b| \leqslant B\gamma q_*^{1/2}$, at an **auxiliary scale** q_*, and $\Theta_q \to 0$ as $q \to \infty$. Bare mass $\mu_N = \lim_{q\to\infty} \mu_q$: an output of the flow.

For q_* large, uniformly in N. Unique solution in a ball, $|\mu_q| \leqslant C\gamma q$, $v_1(\Theta_q) \leqslant C\gamma^2 q^{-1/2}$; increments $\mathcal{N}_k(\ell^{[j]} - \ell^{[j-1]}) \leqslant C_k g_q^{j+1} q\, r_q^{2-k}$; and the defect of $\mathscr{F}^\circ$: $S_4(\mathscr{D}_q^\circ) \leqslant C\gamma^5 q^{-3}$.

Explicit counterterm. $\mu_N = r - 3\lambda \mathrm{Tr}\, C + 6\lambda^2 \int C^3$ is admissible: the measure does not depend on q_*.

Two mass renormalisations

The two divergent self-energies of Appendix A.2 of Lecture 4:

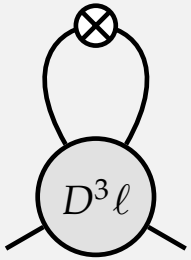
- **Tadpole**: $-\gamma \int_{q_*}^{\infty} \mathrm{Tr} R_{0,q}^2 \, \mathrm{d}q \approx -3\lambda \mathrm{Tr} C \sim -N$. Wick ordering: $\frac{\lambda}{4}\varphi^4 - \frac{3}{2}\lambda c \varphi^2 = \frac{\lambda}{4} H_4(\varphi; c) + \text{const.}$
- **Sunset**: contracting the bubble in $D^3\ell(0)$ with R_0^2 and integrating gives the two-loop self-energy,

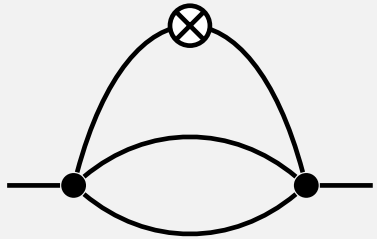
$$6\lambda^2 \int C^3 \approx \frac{3\lambda^2}{8\pi^2} \log N \quad (=2\gamma_T \text{ of Lecture 4, with } \lambda_{L3} = \lambda/4).$$

Power counting: nothing else. In the flow: the integrals from $q = \infty$ of all vertices with $k \geqslant 4$ legs converge; only the linear jet needs a boundary condition.

μ_q at fixed q has a limit as $N \to \infty$; μ_N does not. **Never interchange the two limits.**

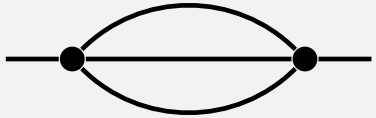
The two mass renormalisations in graphs

$$\partial_q M_q = -\frac{1}{2}$$  $$- \frac{1}{2}$$ 

$$\ni$$ 

tadpole, $D^4U = 2\gamma$

sunset

$$\mu_N \approx -$$  $$+$$ 

$\gamma \operatorname{Tr} C \sim \gamma N$

$\lambda^2 \int C^3 \sim \lambda^2 \log N$

the q-integral removes
the cross: $\int C_q^2\, dq$ closes a line

The cross is $R^2_{0,q}$. The bubble in $D^3\ell^{[2]}(0)$ produces the sunset.

Two correction steps

Backward propagator $\approx s/q$, so $Z_q \approx \int_q^\infty \frac{s}{q} b_s \mathrm{d}s$, and on the ball $|b_s| \lesssim s^{-1} \mathsf{S}_4(\mathscr{D}_s)\, r_s^3$. After n steps, $\mathsf{S}_4(\mathscr{D}_s) \lesssim g_s^{n+1} s^{1/2} r_s^{-4}$:

steps n	$\mathsf{S}_4(\mathscr{D}_s)$	$\lvert b_s\rvert$	$\int_q^\infty \frac{s}{q}\lvert b_s\rvert \mathrm{d}s$
0	$\gamma^3 s^{-2}$	$\gamma^{3/2} s^{-3/2}$	∞ (power)
1	$\gamma^4 s^{-5/2}$	$\gamma^{5/2} s^{-2}$	∞ (log)
2	$\gamma^5 s^{-3}$	$\gamma^{7/2} s^{-5/2}$	$2\gamma^{7/2} q^{-3/2}$

Two steps are the minimum, and suffice. The defect of one step is borderline in the backward equation, as the sunset is borderline in the mass. (Heuristic threshold for Φ^4_3: $n > 1$; the general theory's sufficient criterion is $(n+2)\,\varkappa > (3+\varkappa)/2$, $\varkappa = \frac{1}{2}$, met by $n = 2$.)

The flow at large fields

Lecture 5

The large-field problem

All bounds hold on $\mathcal{B}_q$ only. The free field exceeds r_q with small, nonzero probability, and nothing a priori controls the interacting forecast. We need $\bar{\mathcal{F}}_q$ defined and uniformly coercive for all fields.

Scalar model

$\ell_q(y) = \gamma y[(q+\gamma y^2)^{-1} - q^{-1}]$: power series of radius exactly r_q (singularities at $\pm \mathrm{i} r_q$). Half of the truncations destroy monotonicity. The exact function is harmless, $|\ell_q'| \leqslant \frac{9}{8}\gamma/q$.

Perturbative control for $|y| \lesssim r_q$; outside, a non-perturbative replacement that keeps the structure the estimates use: coercivity, carried by the bare cube.

One site: $q\,y + \lambda y^3 = p$, noise coefficient $H_q(y) = (q+\gamma y^2)^{-1} \leqslant q^{-1} \wedge (\gamma y^2)^{-1}$. Large fields are more elliptic and less noisy.

Clipping

τ smooth odd, $\tau(u)=u$ on $[-1,1]$, constant outside $[-2,2]$, $0\leqslant\tau'\leqslant 1$;

$$\kappa_q(y)=r_q\tau(y/r_q),\quad \tilde{\ell}_q=D\kappa_q\cdot \ell_q^{[2]}\circ\kappa_q,\quad \bar{\mathscr{F}}_q=U'+M_q y+\tilde{\ell}_q.$$

The cubic and linear parts are not clipped, and $D\kappa_q$ keeps the force a gradient. Globally: $s(D^{j-1}\tilde{\ell}_q)\leqslant C g_q q\, r_q^{2-j}$ $(j=2,3,4)$, $s(D\partial_q\tilde{\ell}_q)\leqslant C g_q$.

Global coercivity

For all y: $c(K_q+\gamma y^2)\leqslant H_q(y)^{-1}\leqslant C(K_q+\gamma y^2)$,

$$\|H_q(y)\|_{p\to p}\leqslant \frac{1+C(g_q+g_q^2)}{q},\quad \operatorname{diag} H_q(y)^2\leqslant C a_2(q).$$

Neumann series for $H=(I+G(M+D\tilde{\ell}))^{-1}G$, $G=(K_q+\gamma y^2)^{-1}\leqslant C_q$.

The complete defect

$b = b^{\mathrm{b}} + b^{\mathrm{l}}$, $b^{\mathrm{b}} = -H\, D\kappa\, \mathscr{D}^{\circ}(\kappa y)$, $t_q(y) = y\, 1_{|y|>r_q}$.

Proposition. (uniformly in y, N, p)

$$\|b_q^{\mathrm{b}}\|_p \leqslant C\gamma^{7/2} q^{-5/2}, \qquad \|b_q^{\mathrm{l}}\|_p \leqslant C\gamma\, q^{-3/2}(\|t_q\|_p + r_q^{-1}\|y^2 1_{|y|>r_q}\|_p),$$

$$\|b_q^{\mathrm{l}}\|_2^2 \leqslant C\gamma^4 q^{-5}\|t_q\|_6^6.$$

Sources of b^{l} (four groups): the mass term $M'y - JM'\kappa y$ ($J = D\kappa$, $M' = \partial_q M$; linear part unclipped, flow computed at κy); the motion of the clip; cubic vertex at y vs correction at κy; chain-rule terms with κ'', κ'''. All vanish if $\|y\|_\infty \leqslant r_q$.

The point

b^{l} is as large as the whole drift, but **supported on large fields**. It is controlled by how often the actual dynamics visits them: $\int q^{-3/2}\gamma \|t_q(Y_q)\|_6^6\, \mathrm{d}q$ (Lecture 6).

The coordinate equation

With $DU = \frac{\gamma}{3} y^3$:

$$K_q \Phi_q = K_q(\Upsilon_q + \varrho_q(\Upsilon_q)) + \frac{\gamma}{3}\Upsilon_q^3, \qquad \varrho_q(y) = C_q(M_q y + \tilde{\ell}_q(y)), \quad \|D\varrho_q\|_{p\to p} \leqslant C(g_q + g_q^2).$$

Up to a small Lipschitz perturbation, the forecast solves the cubic elliptic equation $K_q(\Upsilon + \varrho) + \frac{\gamma}{3}\Upsilon^3 = K_q \Phi_q$ driven by the field at resolution q.

Summary of today, at every lattice:

- μ_N, M_q and a globally coercive $\bar{\mathscr{F}}_q$, all deterministic;
- the exact FBSDE for (Υ, Z);
- a source whose bounded-field part is integrable (two steps), and whose large-field part is controlled by sixth powers of Υ on large fields.

Next: estimates on (Υ, Z) uniform in N.

The force below the auxiliary scale

The expansion needs g_q small, so $\bar{\mathscr{F}}_q$ is built for $q \geqslant q_*$. Below q_* nothing needs constructing: freeze it,

$$\bar{\mathscr{F}}_q(y) = \bar{\mathscr{F}}_{q_*}(y) + (q_* - q)y.$$

Then $K_q + D\bar{\mathscr{F}}_q = K_{q_*} + D\bar{\mathscr{F}}_{q_*}$: coercivity is automatic, $H_q = H_{q_*}$, and $y_q(x) = y_{q_*}(C_{q_*}K_q x)$.

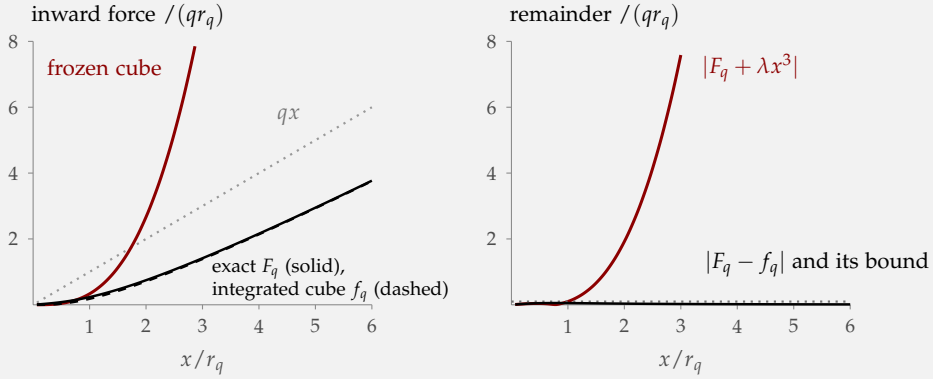
The defect is explicit: $\mathscr{D}_s(y) = -y + \frac{1}{2}\mathsf{C}\mathrm{tr}_{H^2_{q_*}} D^2\bar{\mathscr{F}}_{q_*}(y)$, with $\|\mathscr{D}_s(y)\|_p \leqslant C\|y\|_p$.

So the system holds on $[q_0, \infty)$ with $\Upsilon_{q_0} = 0$: the initial value is deterministic, with no large mass. q_* stays only in the proofs, where the expansion and the closure of the loop need g_q small.

Why not Polchinski with a frozen cube?

Programme: keep $-\lambda\varphi^3$ unclipped, clip the higher orders, pay in the source of $R = F - f$. Test it on one site: $K_q = q$, $S = 1/q$.

$$F_q(x) = -\lambda \mathbb{E}_x Y^3 = q(m_q(x) - x), \quad |m_q(x)| \leqslant 2\lambda^{-1/4} + 2(q|x|/\lambda)^{1/3}.$$



So $R/(\lambda x^3) \to 1$: the remainder **cancels the cube**. Field equation: $qx = (q - q_0)m_q + \eta_q$, only linear confinement.

The price of clipping

$f^c = f_1 + \mathscr{C} f_2$, with $f_1 = -\lambda(x^3 + 3S x)$, f_2 the first tree, $\mathscr{C} f_2 = \kappa_q' \cdot f_2 \circ \kappa_q$. For $|x| \geqslant 2r_q$, the Burgers defect is exactly

$$E_q = S^2 f_1 f_1' \approx 3\lambda^2 S^2 x^5,$$

of size $(q/\lambda)^{1/2}$ at the edge of the clip: not integrable in q.

$$\mathrm{d}R_q = -[E_q + S^2(f^c)' R_q]\,\mathrm{d}q + \Xi_q\,\mathrm{d}W_q, \quad \text{bracket} \approx 3\lambda^2 S^2 x^5 - 3\lambda S^2 x^2 R.$$

The source and the dissipation cancel for the true $R \sim \lambda x^3$: separate estimates give nothing.

Tails independent of R: on one site from the posterior moment, but not uniform on the lattice (negative bare mass).

More orders do not help: the tree series has radius $\frac{2}{3} r_q$.

Integrating the cube

Solve the transport part of the flow exactly:

$$\partial_q f + S^2 f f' = 0, \quad f_\infty = -\lambda x^3 \quad \Rightarrow \quad f_q(x) = -\lambda y^3, \quad q\,y + \lambda y^3 = q\,x.$$

$$\sup_x |F_q(x) - f_q(x)| \leqslant 2C_*(\lambda/q)^{1/2}, \quad C_* < \frac{1}{2}.$$

- $y_q(\Phi_q) = Y_q$: **this is the forecast coordinate**. $F_q - f_q = K_q Z_q$, and Z is the posterior mean minus the mode.
- The defect is $q\,b_q(y)$: bounded for all fields. The trees are in $x \mapsto y$; the cube does not flow in y, so only loops are clipped.
- The cube is coercive in y: $q\,y + \lambda y^3 = q\,\Phi_q$.

Not a no-go theorem: it rules out only estimating R as a small perturbation of the frozen cube. Integrating the cube in the original field is possible, but on the lattice its loop defect must still be renormalised; that route is not proved here.